

DIGITAL IMAGE PROCESSING

LECTURE # 9

IMAGE ENHANCEMENT IN FREQUENCY DOMAIN

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Topics to Cover

- Fourier Series
- Fourier Transform
- Discrete Fourier Transform
- Low Pass Filtering
- High Pass Filtering
- Band Pass Filtering
- Low Pass Filters
 - ❑ Ideal Low Pass Filter
 - ❑ Butterworth Low Pass Filter
 - ❑ Gaussian Low Pass Filter
- High Pass Filters
 - ❑ Ideal High Pass Filter
 - ❑ Butterworth High Pass Filter
 - ❑ Gaussian High Pass Filter

Joseph Fourier: History [2]

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- *Jean Baptiste Joseph Fourier*, French mathematician and physicist (03/21/1768-05/16/1830)

Orphaned: at age 8

(The Analytic Theory of Heat)



Permanent Secretary of
the French Academy of
Sciences: 1822

In 1830, he was elected
a foreign member of the
Royal Swedish Academy
of Sciences.

Background (Fourier Series & Fourier Transform)

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- Any function that periodically repeats itself can be expressed as the sum of sine's and cosines of different frequencies each multiplied by a different coefficient
- This sum is known as Fourier Series
- It does not matter how complicated the function is; as long as it is periodic it can be represented by such a sum
- Even functions that are not periodic can be expressed as the integrals of sine's and cosines multiplied by a weighing function
- This is known as Fourier Transform

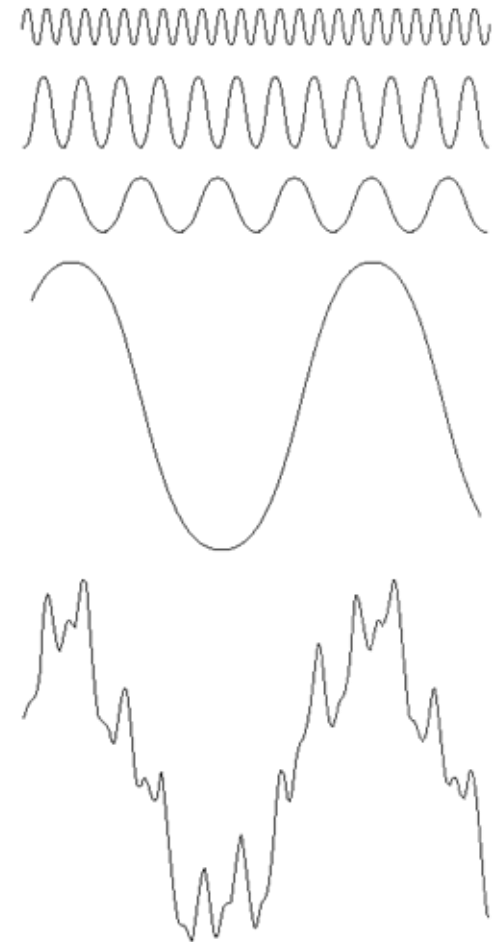
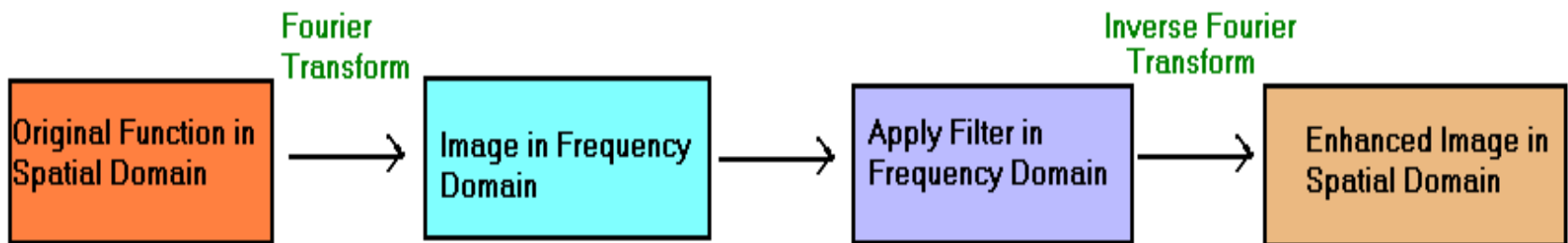


FIGURE 4.1 The function at the bottom is the sum of the four functions above it. Fourier's idea in 1807 that periodic functions could be represented as a weighted sum of sines and cosines was met with skepticism.

Background (Fourier Series & Fourier Transform)

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- ❑ A function expressed in either a Fourier Series or transform can be reconstructed completely via an inverse process with no loss of information
- ❑ This is one of the important characteristics of these representations because they allow us to work in the Fourier Domain and then return to the original domain of the function
- ❑ The original function is often a function in the *Time domain*
- ❑ In image Processing the original function is in the *Spatial Domain*



One-Dimensional Fourier Transform and its Inverse

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- The Fourier transform $F(u)$ of a single variable, continuous function $f(x)$ is

$$F(u) = \int_{-\infty}^{\infty} f(x) e^{-j2\pi ux} dx \quad \text{where } j = \sqrt{-1}.$$

- Given $F(u)$ we can obtain $f(x)$ by means of the Inverse Fourier Transform

$$f(x) = \int_{-\infty}^{\infty} F(u) e^{j2\pi ux} du.$$

- (u) is the frequency variable.
- $F(u)$ is composed of an infinite sum of sine and cosine terms and...
- Each value of u determines the frequency of its corresponding sine-cosine pair.

Discrete Fourier Transforms (DFT)

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- 1-D DFT for M samples is given as

$$F(u) = \frac{1}{M} \sum_{x=0}^{M-1} f(x) e^{-j2\pi ux/M} \text{ for } u = 0, 1, 2, \dots, M-1$$

- The inverse Fourier transform in 1-D is given as

$$f(x) = \sum_{u=0}^{M-1} F(u) e^{j2\pi ux/M} \text{ for } x = 0, 1, 2, \dots, M-1$$

Since $e^{j\theta} = \cos \theta + j \sin \theta$, Euler's formula

$$F(u) = \frac{1}{M} \sum_{x=0}^{M-1} f(x) [\cos(2\pi ux/M) - j \sin(2\pi ux/M)].$$

$$F(u) = \underbrace{R(u)}_{\text{Real}} + j \underbrace{I(u)}_{\text{Imaginary}}$$

Two-Dimensional Fourier Transform and its Inverse

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- The Fourier transform $F(u,v)$ of a two variable, continuous function $f(x,y)$ is

$$F(u, v) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) e^{-j2\pi(ux+vy)} dx dy.$$

- Given $F(u,v)$, we can obtain $f(x,y)$ by means of the Inverse Fourier Transform

$$f(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F(u, v) e^{j2\pi(ux+vy)} du dv.$$

2D DFT

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2-D DFT for a total of $M \times N$ samples is given as

$$F(u, v) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)}$$

for $u = 0, 1, 2, \dots, M-1$ and $v = 0, 1, 2, \dots, N-1$

2-D inverse DFT is given as

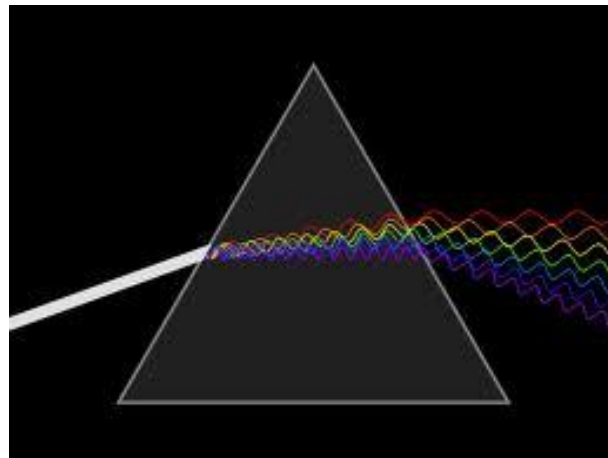
$$f(x, y) = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi(ux/M + vy/N)}$$

for $x = 0, 1, 2, \dots, M-1$ and $y = 0, 1, 2, \dots, N-1$

Fourier Transform

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- ❑ A useful analogy is to compare the Fourier Transform to a glass prism
- ❑ The Prism is a physical device that separates light into various color components depending upon its wavelength or frequency content
- ❑ The Fourier Transform may be viewed as a “mathematical prism” that separates a function into various components also based on frequency content



2D DFT

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2D DFT pair $f(x,y) \longleftrightarrow F(u,v)$

$$F(u,v) = R(u,v) + jI(u,v)$$

real part

imaginary part

(Fourier)
Magnitude spectrum

$$|F(u,v)| = \sqrt{R^2(u,v) + I^2(u,v)}$$

(Fourier)
Phase spectrum

$$\phi(u,v) = \angle F(u,v) = \tan^{-1} \left[\frac{I(u,v)}{R(u,v)} \right]$$

Power spectrum

$$P(u,v) = |F(u,v)|^2$$

Filtering using Fourier Transforms

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- Multiplication in the frequency domain is equivalent to convolution in the spatial domain

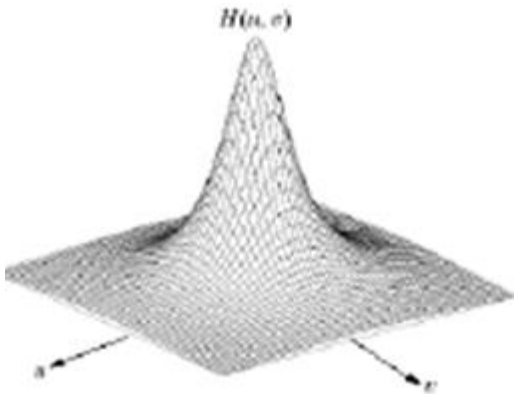
$$g(x, y) = f(x, y) * h(x, y) \Leftrightarrow F(u, v) \cdot H(u, v) = G(u, v)$$

- $H(u, v)$ plays an important role $\Rightarrow$ determine how the enhanced image looks like
- A lowpass $H(u, v)$ results in a *blurring* effect; a highpass $H(u, v)$ results in a *sharpening* effect
- **High frequency**: edges and sharp details in an image
- **Low frequency**: slowly varying characteristics of an image (e.g. overall contrast and average intensity)

Low Pass Filtering

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- Regions of relatively uniform gray values in an image contribute to low-frequency content of its Fourier transform.
- A low-pass filter attenuates high frequencies and retains low frequencies unchanged.
- Hence, an image can be smoothed in the Frequency domain by attenuating the high-frequency content of its Fourier transform. This would be a low pass filter!
- The result in the spatial domain is equivalent to that of a **smoothing filter**; as the blocked high frequencies correspond to sharp intensity changes



Original Image

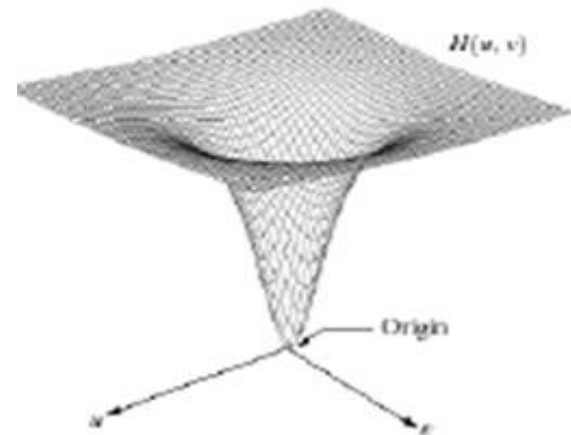


LPF image, $r_0 = 57$

High Pass Filtering

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- Edges and sharp transitions in gray values in an image contribute significantly to high-frequency content of its Fourier transform.
- Hence, image sharpening in the Frequency domain can be done by attenuating the low-frequency content of its Fourier transform. This would be a high pass filter!
- A high pass filter, yields edge enhancement or edge detection in the spatial domain, because edges contain many high frequencies.
- Areas of rather constant gray level consist of mainly low frequencies and are therefore suppressed.



Band Pass Filtering

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- ❑ A band-pass attenuates very low and very high frequencies, but retains a middle range band of frequencies.
- ❑ Band-pass filtering can be used to enhance edges (suppressing low frequencies) while reducing the noise at the same time (attenuating high frequencies).
- ❑ Band-pass filters are a combination of both low pass and high pass filters. They attenuate all frequencies smaller than a frequency D_0 and higher than a frequency D_1 , while the frequencies between the two cut-offs remain in the resulting output image.

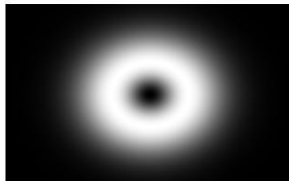
Original Image



Fourier Spectrum of Image



Frequency Domain Filter Function Image



Bandpass Filtered Image



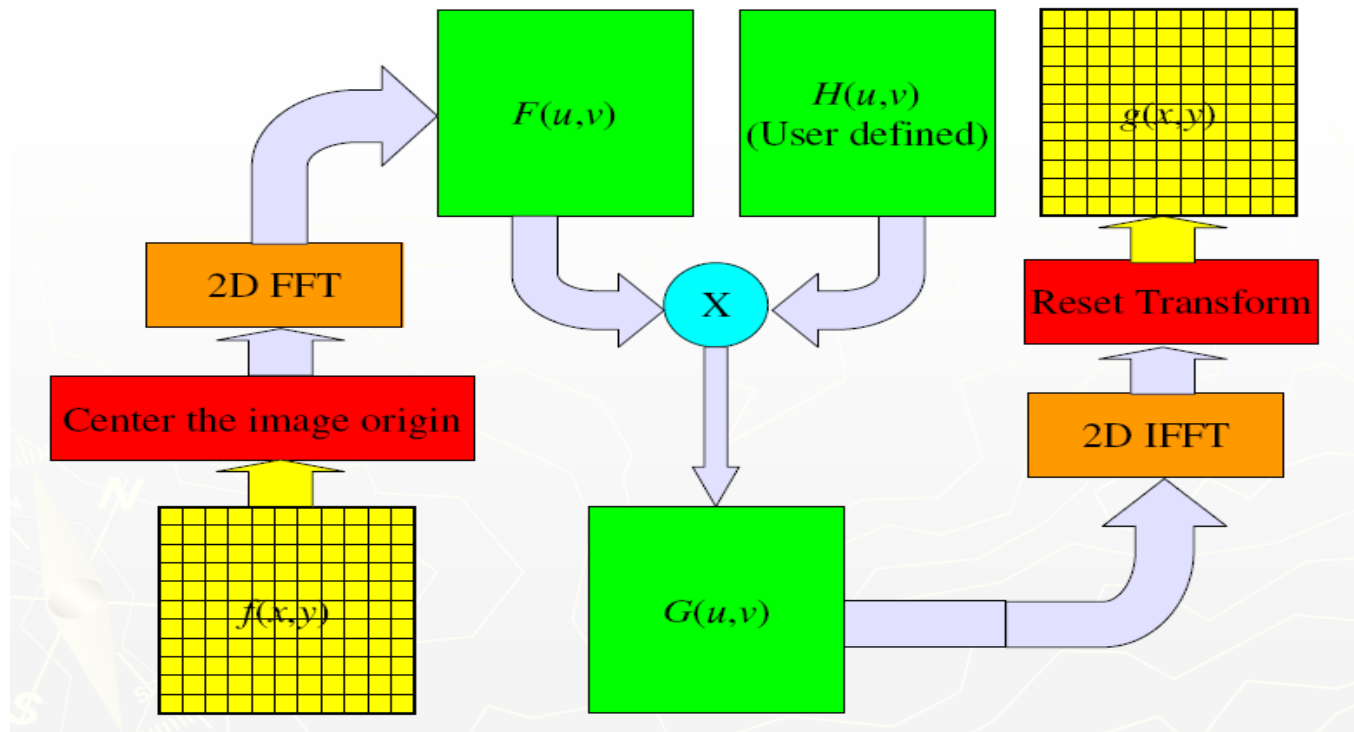
Effect of Different Cutoff Frequencies

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- ❑ As Cutoff frequencies decreases
 - ✓ Image becomes more blurred
 - ✓ Noise Reduces
 - ✓ Analogous to larger Spatial Filter sizes

Image Enhancement in Frequency Domain

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$$g(x, y) = h(x, y) \otimes f(x, y) \Leftrightarrow G(u, v) = H(u, v)F(u, v)$$
$$h(x, y) = IDFT(H(u, v)), \quad H(u, v) = DFT(h(x, y)).$$

Shifting the Origin to the Center

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If we put $u_0 = M / 2$, and $v_0 = N / 2$,

$$\begin{aligned} e^{j 2 \pi (u_0 x / M + v_0 y / N)} &= e^{j \pi (x+y)} \\ &= (-1)^{(x+y)} \end{aligned}$$

This means that

$$f(x, y)(-1)^{(x+y)} \Leftrightarrow F(u - M / 2, v - N / 2)$$

To transform the origin to the center in the transformed image the input is always multiplied by the factor $(-1)^{(x+y)}$

Ideal Low Pass Filter

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- The simplest lowpass filter is ILPF, it (cuts off) all high frequency components that are at distance greater than a specified distance: D_0 from the center

$$H(u, v) = \begin{cases} 1 & \text{if } D(u, v) \leq D_0 \\ 0 & \text{if } D(u, v) > D_0 \end{cases}$$

Where

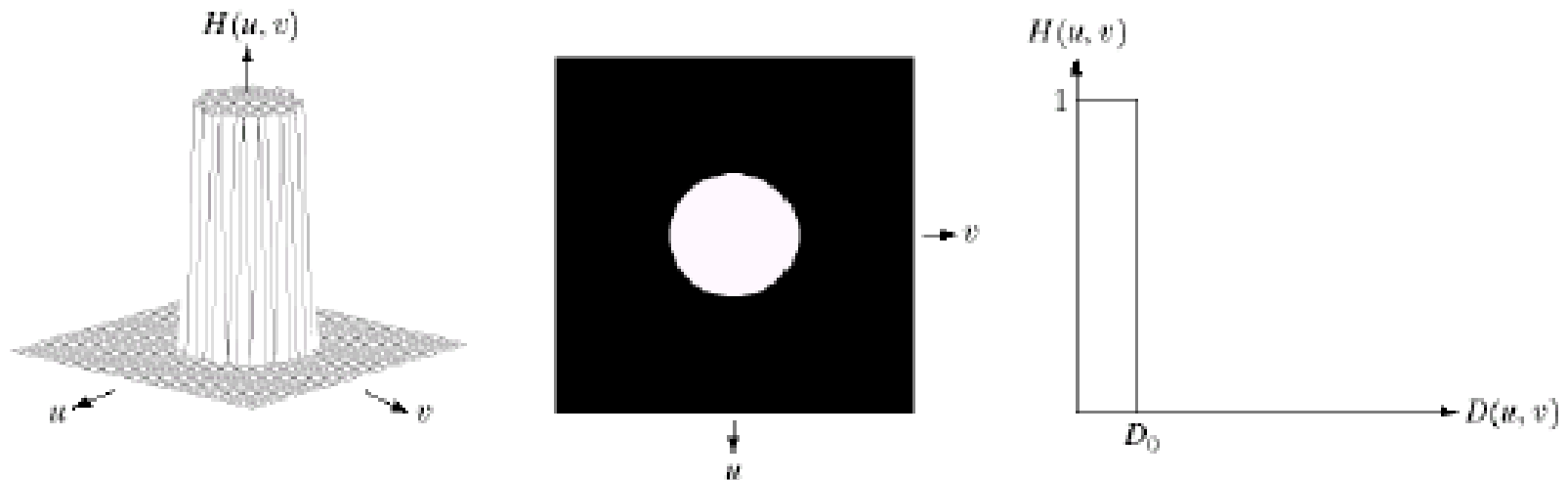
- ✓ $D(u, v)$ is the distance from (u, v) to the center
- ✓ Image size: $M \times N$
- ✓ Center of the frequency rectangle: $(u, v) = (M/2, N/2)$
- ✓ Distance to the center: $D(u, v) = [(u - M/2)^2 + (v - N/2)^2]^{1/2}$

Ideal Low Pass Filter

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$$H(u, v) = \begin{cases} 1, & \text{if } D(u, v) \leq D_0 \\ 0, & \text{if } D(u, v) > D_0 \end{cases} \quad D(u, v) = \sqrt{\left(u - \frac{M}{2}\right)^2 + \left(v - \frac{N}{2}\right)^2}$$

D_0 : cutoff frequency



a b c

FIGURE 4.10 (a) Perspective plot of an ideal lowpass filter transfer function. (b) Filter displayed as an image. (c) Filter radial cross section.

Low Pass Filtering

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- ❑ The Cutoff frequency of the Ideal LPF determines the amount of frequency components passed by the filter
- ❑ Smaller the value of D_0 , more the number of image components eliminated by the filter
- ❑ In general, the value of D_0 is chosen such that most components of interest are passed through, while most components not of interest are eliminated.



Original Image



LPF image, $r_0 = 57$



LPF image, $r_0 = 36$



LPF image, $r_0 = 26$

Ideal Low Pass Filter

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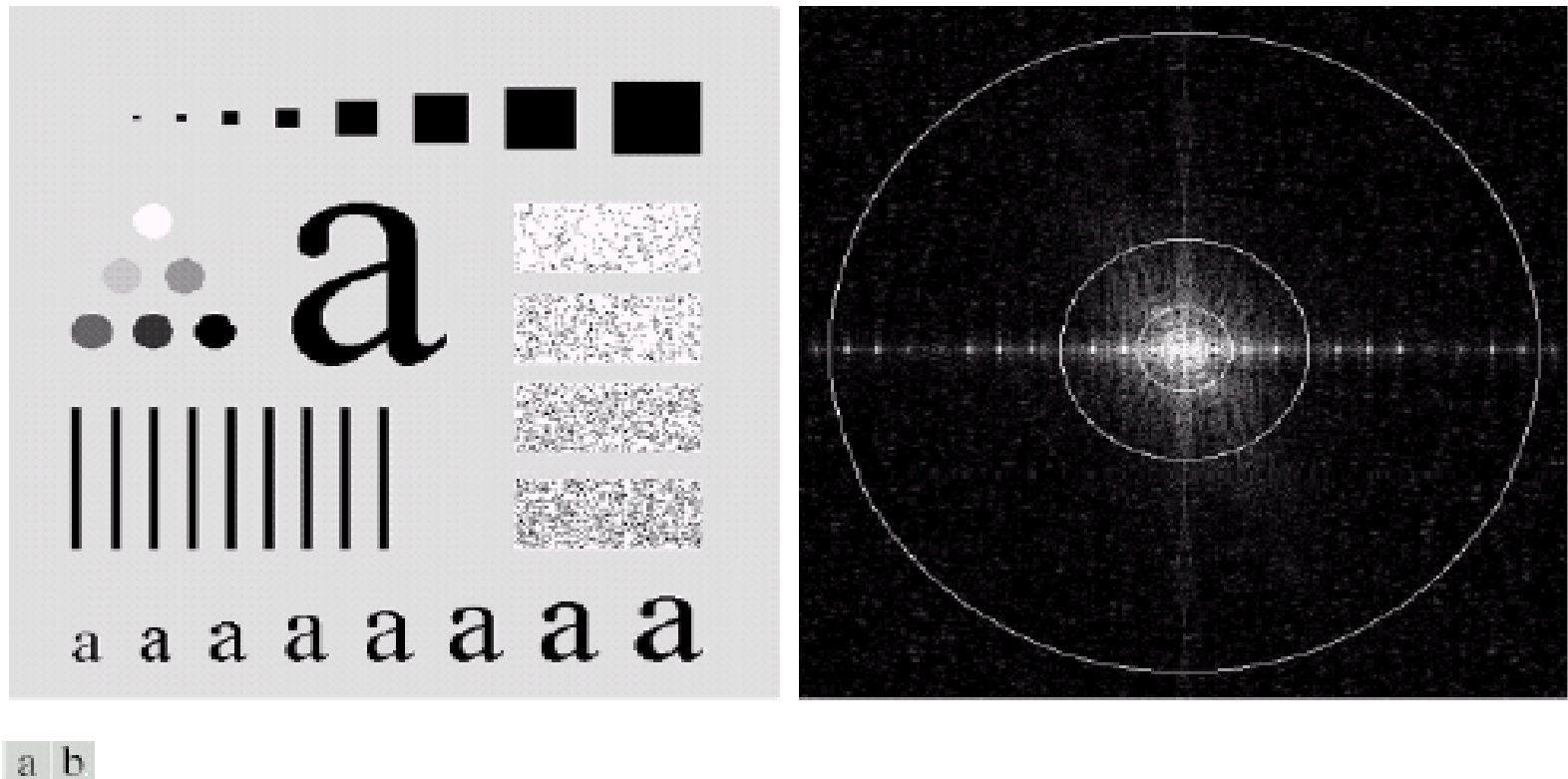


FIGURE 4.11 (a) An image of size 500×500 pixels and (b) its Fourier spectrum. The superimposed circles have radii values of 5, 15, 30, 80, and 230, which enclose 92.0, 94.6, 96.4, 98.0, and 99.5% of the image power, respectively.

Ideal Low Pass Filter

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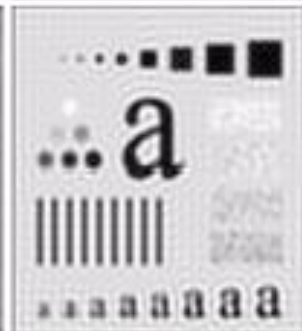
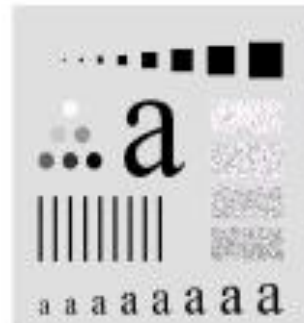
Note that the narrower the filter in the freq. domain is the more severe are the blurring and ringing!

original

15 pixels
94.6%

Ringing effect
due to ideal filter
behavior

80 pixels
98.9%



5 pixels
92.0%
Useless,
severe blurring
(8% → detail)

30 pixels
96.4%
Ringing effect
due to ideal filter
behavior

230 pixels
99.5%

Why ringing appears in ILPF?

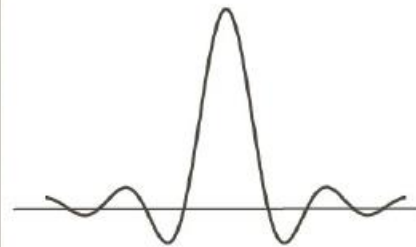
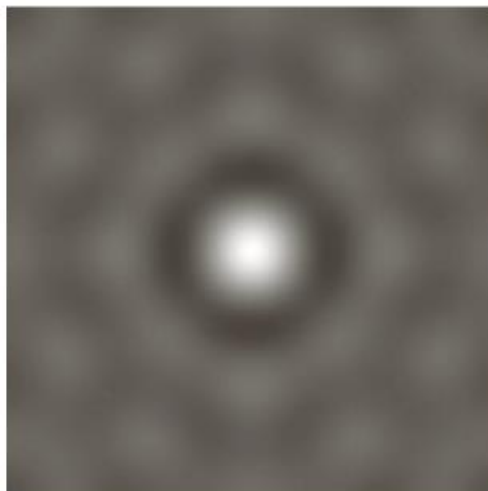
25



Ringling & Blurring Features of ILPF

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- The ILPF has **sharp** cutoffs or discontinuities which cause ringing.
- ✓ The ILPF has a sink function behavior in the SD.
- ✓ The center of the lobe is the cause for blurring, **but** the outer smaller lobes cause ringing.
- ✓ We want to achieve blurring with **little** ringing.



a b

FIGURE 4.43
(a) Representation in the spatial domain of an ILPF of radius 5 and size 1000×1000 .
(b) Intensity profile of a horizontal line passing through the center of the image.

Butterworth Low Pass Filter

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Provide a smooth transition between low and high frequencies

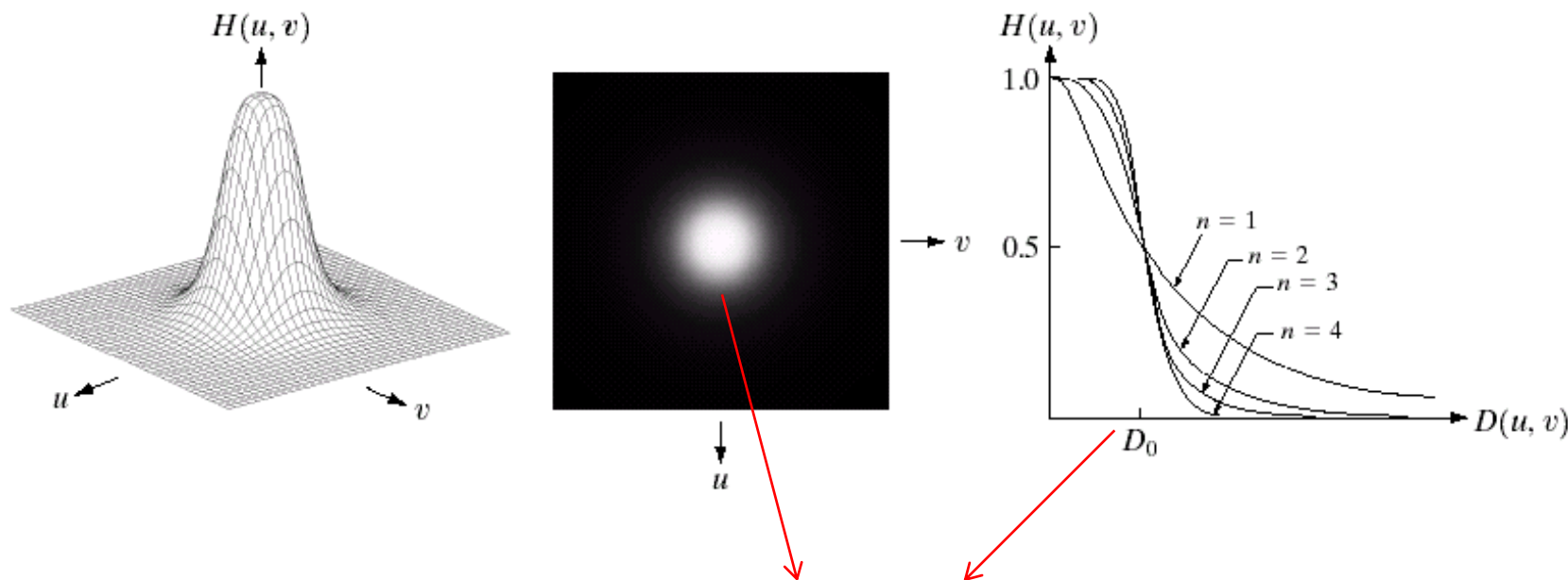
Reduce the ringing effect caused by the ILPF

Transfer function of Butterworth LPF (order n , cutoff frequency D_0):

$$H(u, v) = \frac{1}{1 + \left[\frac{D(u, v)}{D_0} \right]^{2n}} = \frac{1}{1 + \left[\frac{(u - M/2)^2 + (v - N/2)^2}{D_0^2} \right]^n}$$

Butterworth Low Pass Filter

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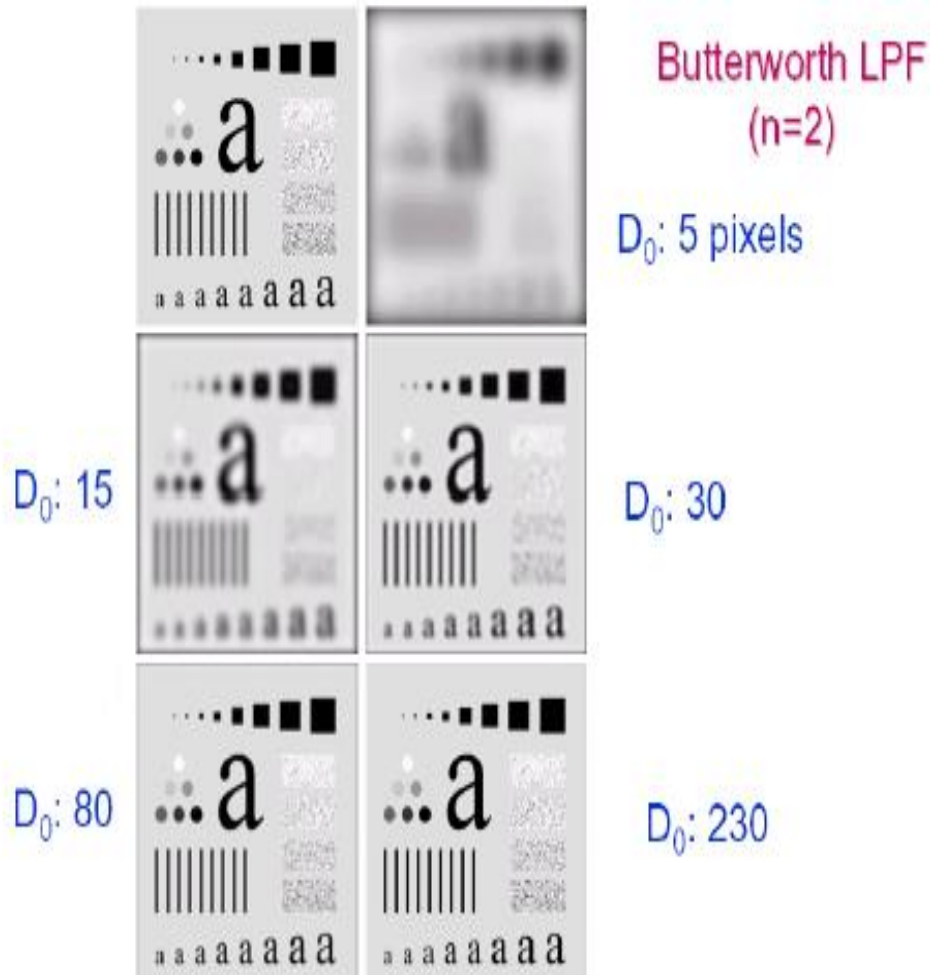


Note:

➤ $H(U, V) = 0.50$ (50% from its maximum value of 1) when $D(u, v) = D_0$

Butterworth Low Pass Filter

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- Less ringing effect due to the filter's smooth transition between low and high frequencies
- A Butterworth filter of order 1 has no ringing
- Ringing is generally imperceptible in filters of order 2, but becomes significant for higher orders
- A filter of order 2 is a good compromise between effective lowpass filtering and acceptable ringing behavior

Butterworth Low Pass Filter

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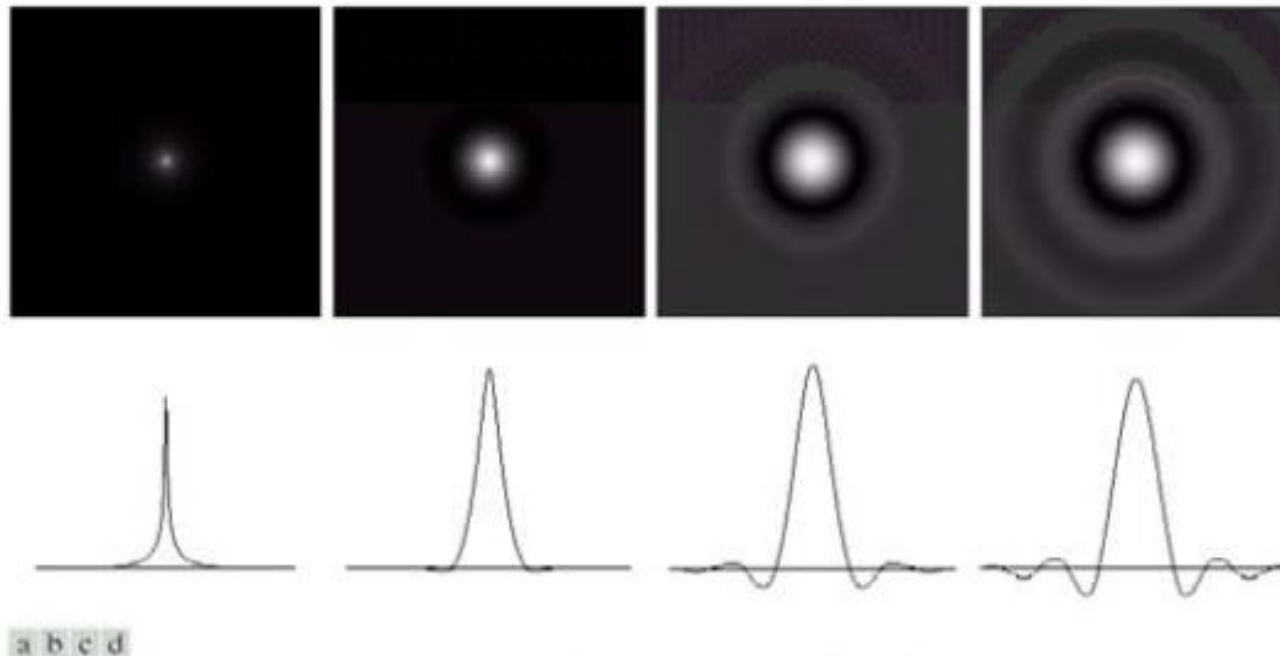


FIGURE 4.16 (a)–(d) Spatial representation of BLPFs of order 1, 2, 5, and 20, and corresponding gray-level profiles through the center of the filters (all filters have a cutoff frequency of 5). Note that ringing increases as a function of filter order.

no ringing for $n=1$, imperceptible ringing for $n=2$, ringing increases for higher orders (getting closer to Ideal LPF).

Filter of order 20 approximates the ringing behavior of ideal LPF

Gaussian Low Pass Filter

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- The transfer function of a Gaussian lowpass filter is defined as

$$H(u, v) = e^{-D^2(u, v) / 2D_0^2}$$

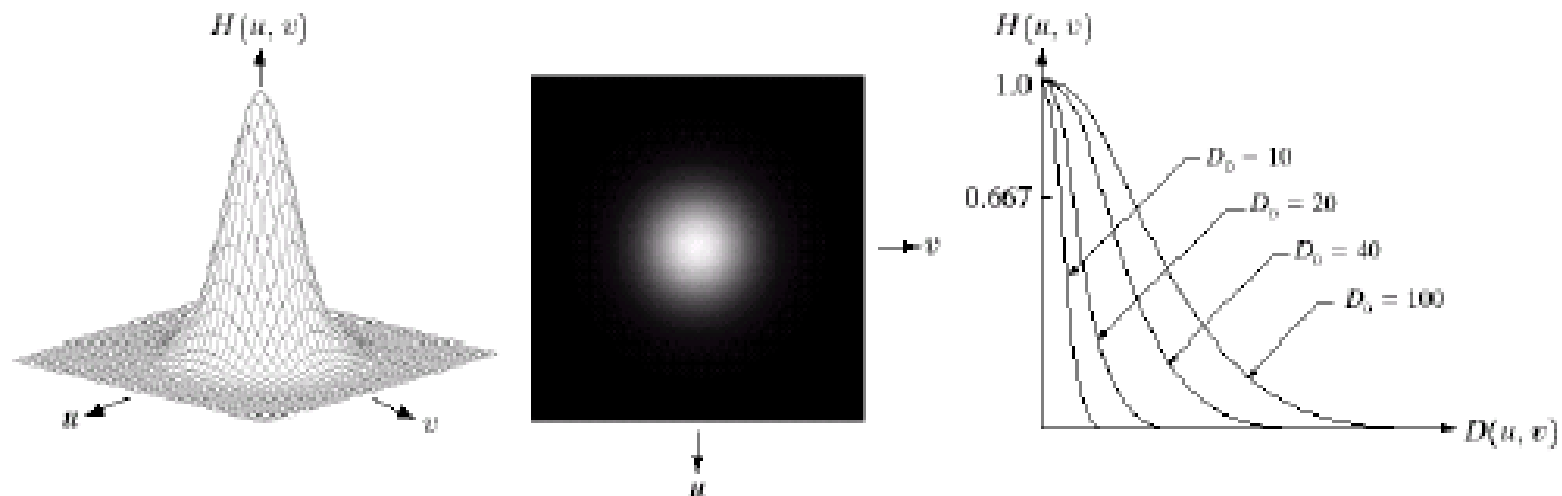
$$H(u, v) = e^{-\frac{D^2(u, v)}{2\sigma^2}} \xrightarrow{\sigma=D_0} H(u, v) = e^{-\frac{D^2(u, v)}{2D_0^2}}$$

$$H(D(u, v) = 0) = 1 \qquad D^2(u, v) = \left(u - \frac{M}{2}\right)^2 + \left(v - \frac{N}{2}\right)^2$$

$$H(D(u, v) = D_0) = e^{-0.5} = 0.607$$

Gaussian Low Pass Filter

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a b c

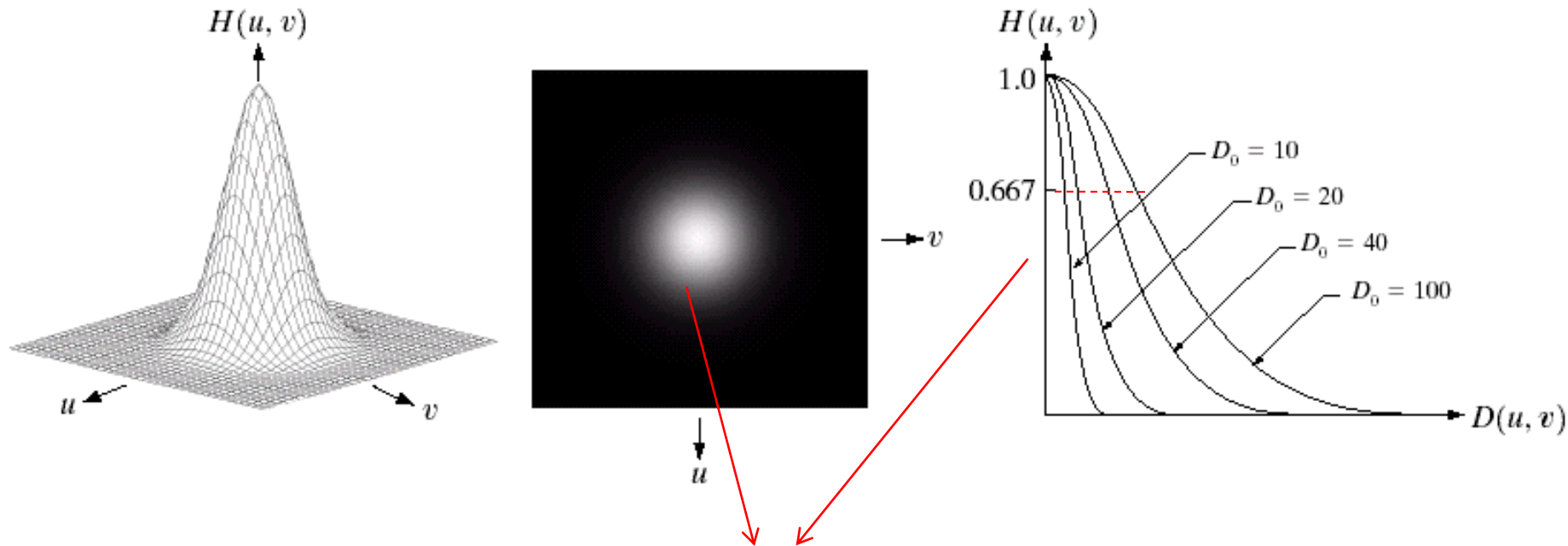
FIGURE 4.17 (a) Perspective plot of a GLPF transfer function. (b) Filter displayed as an image. (c) Filter radial cross sections for various values of D_0 .

Note:

The IFT (INVERSE FOURIER) of Gaussian is also Gaussian. Hence there is **no ringing** effect.

Gaussian Low Pass Filter

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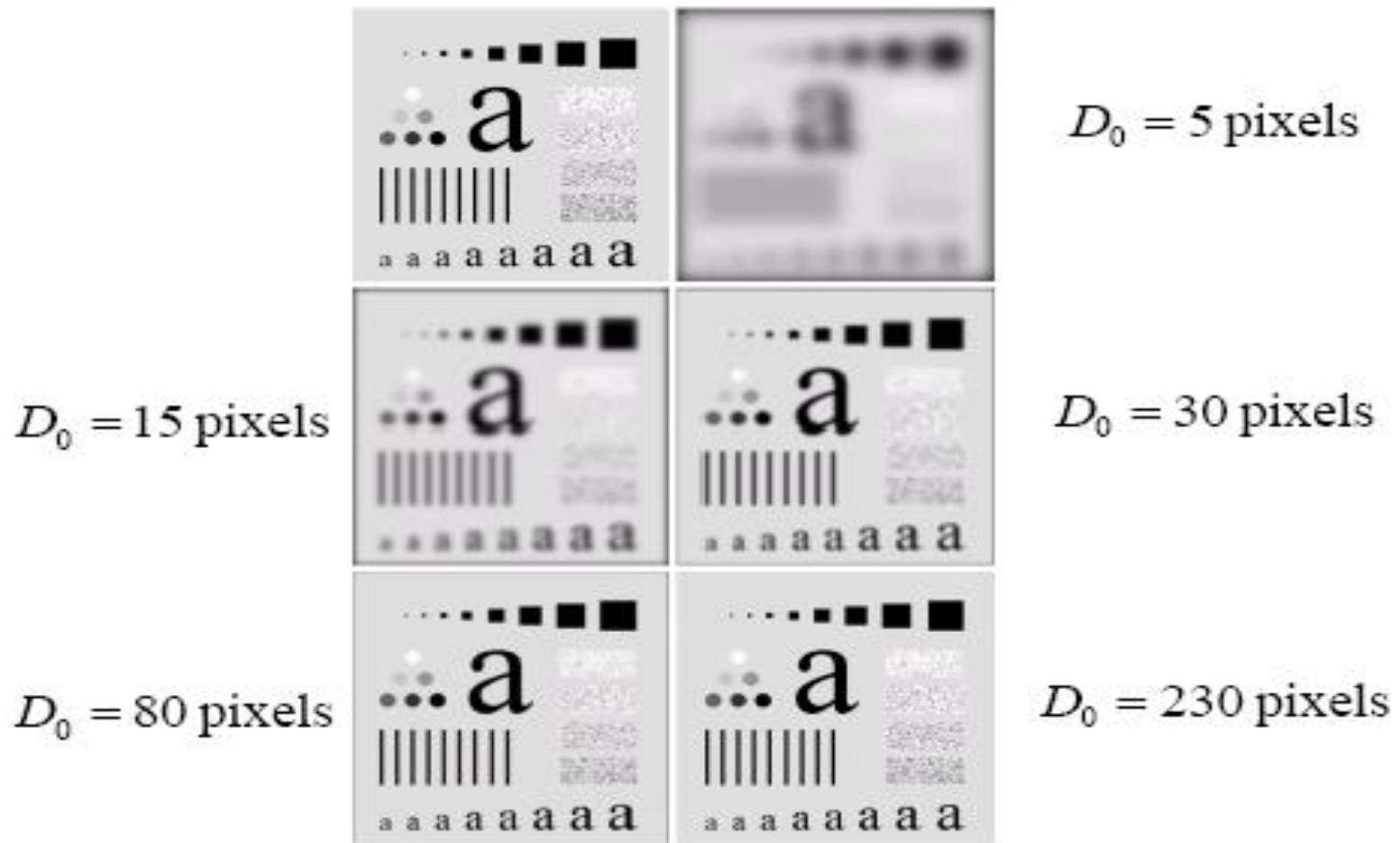


Note:

➤ $H(u, v) = 0.667$ when $D(u, v) = D_0$

Gaussian Low Pass Filter

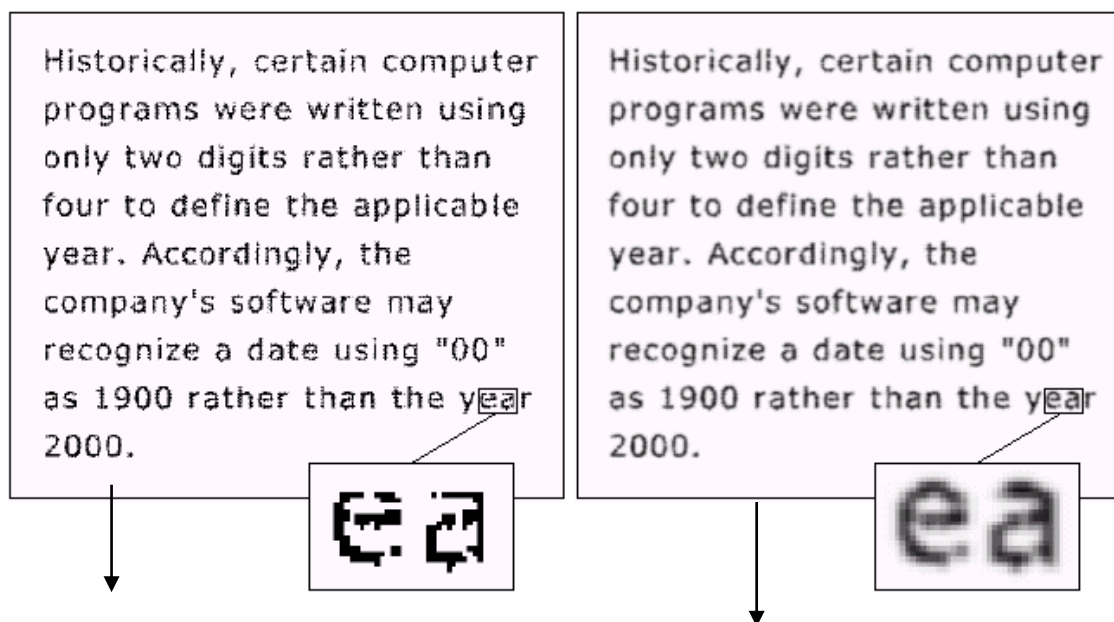
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Gaussian Low Pass Filter

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- A low pass Gaussian filter is used to connect broken text



Machine recognition systems have difficulty in reading broken characters.

GLPF with $D0 = 80$

Gaussian Low Pass Filter

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- Different low pass Gaussian filters used to remove blemishes in a photograph



How does D_0 control smoothing?

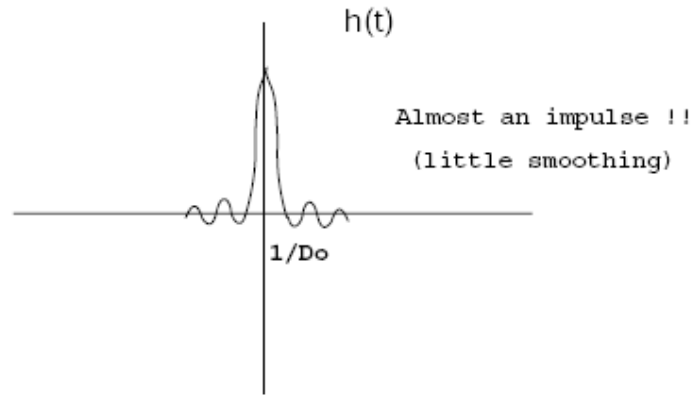
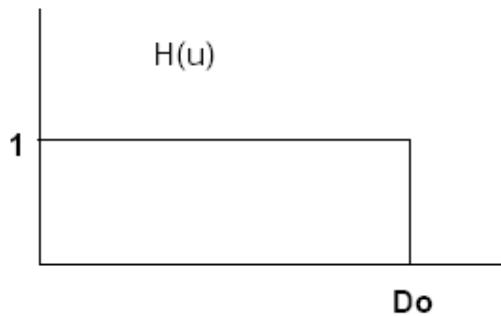
37



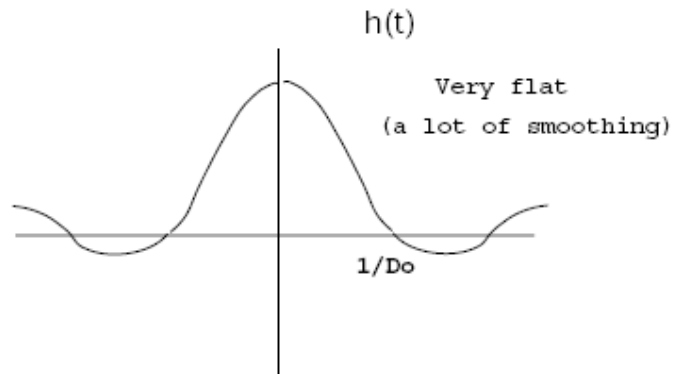
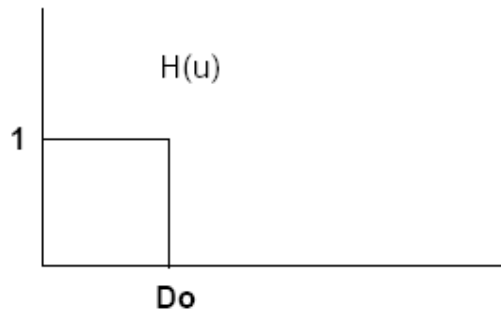
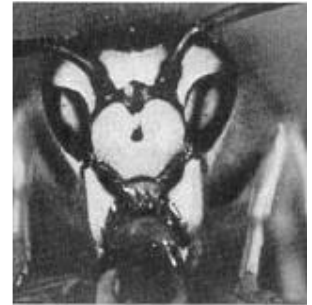
How does D_0 control smoothing?

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- D_0 controls the amount of blurring



$r=78$ (99%)



$r=8$ (90%)



Image Sharpening in Frequency Domain

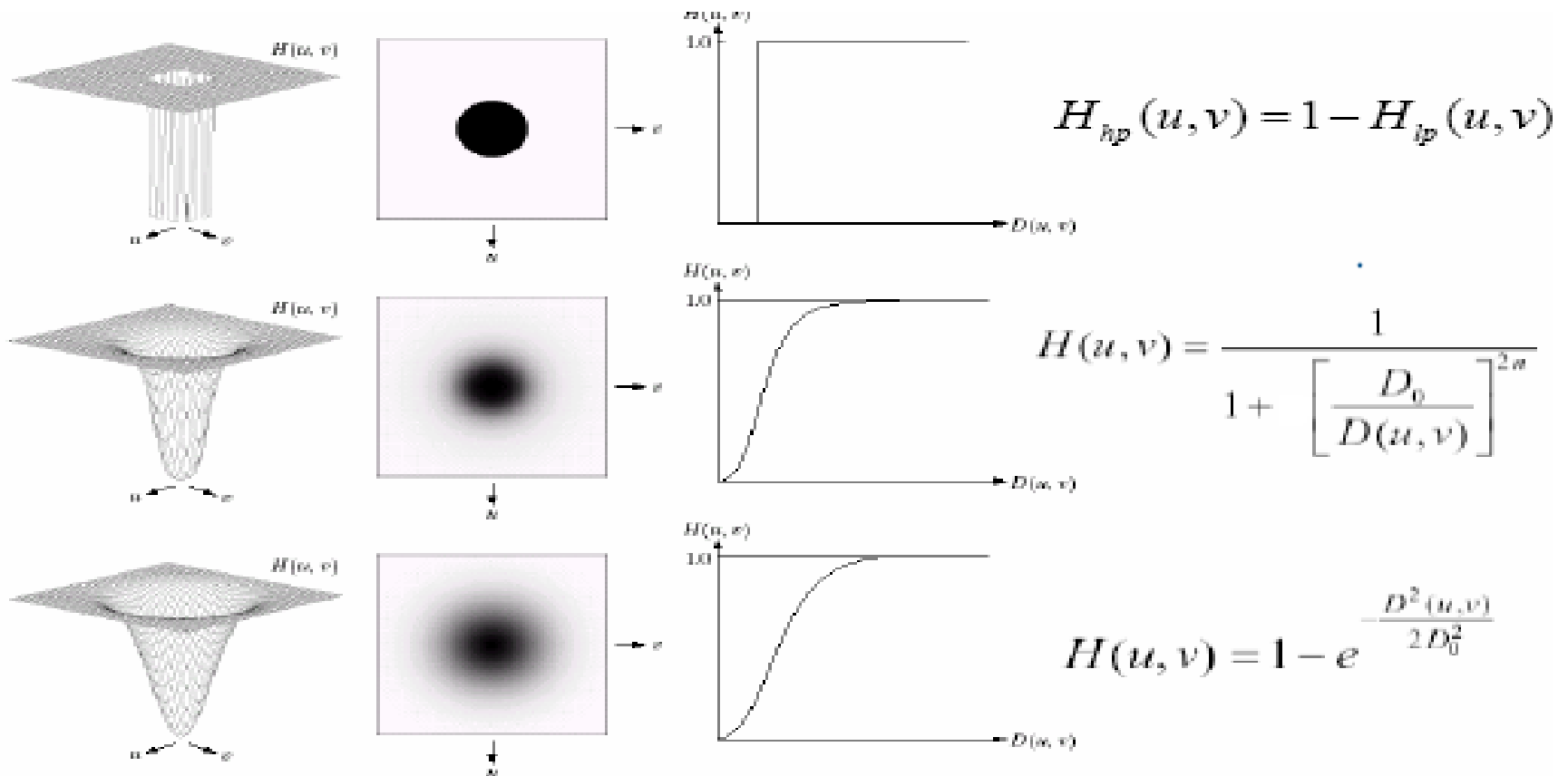
39

- ❑ Blurring (smoothing) is achieved by attenuating the **HF(high frequency)** components of DFT of an image. (low pass filters)
- ❑ Sharpening is achieved by attenuating the **LF (low frequency)** components of DFT of an image. Where edges and fine detail in images are associated with high frequency components (high pass filters)
- ❑ **High pass filters** – only pass the high frequencies, drop the low ones. High pass frequencies are precisely the inverse of low pass filters, so:

$$H_{hp}(u, v) = 1 - H_{lp}(u, v)$$

Sharpening Fourier Domain Filters

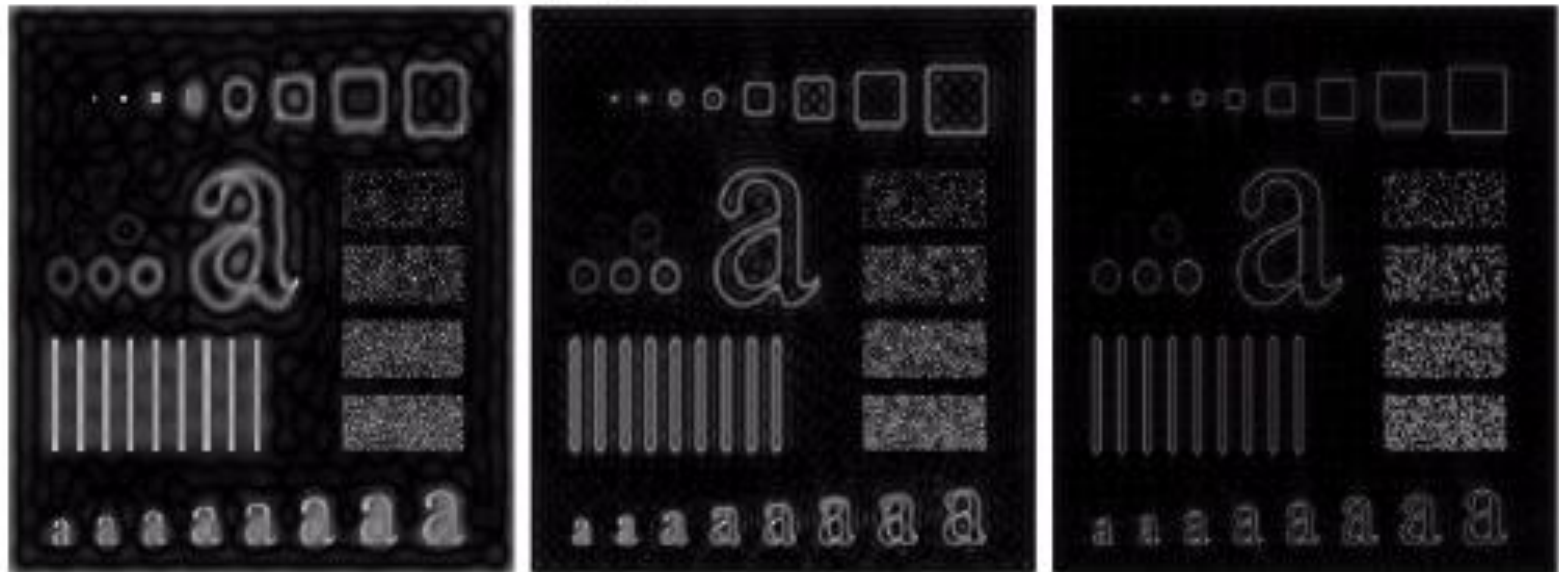
40



Sharpening Fourier Domain Filters

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Results of Ideal High pass Filters



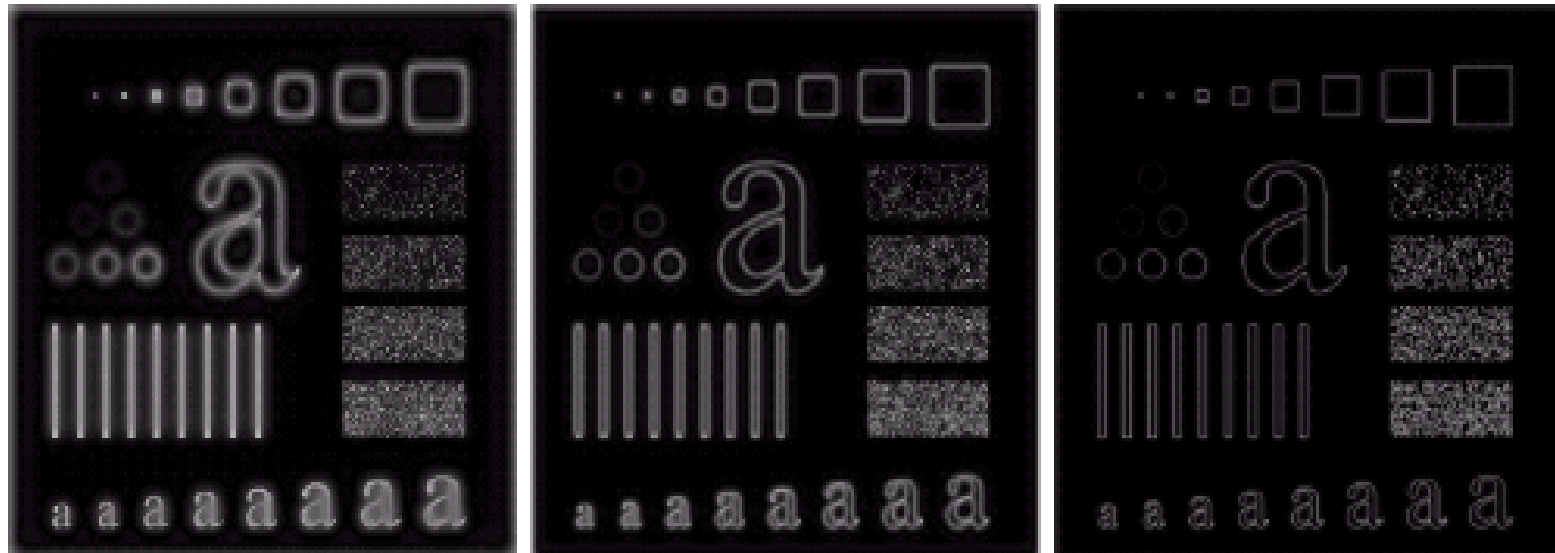
a b c

FIGURE 4.24 Results of ideal highpass filtering the image in Fig. 4.11(a) with $D_0 = 15, 30$, and 80 , respectively. Problems with ringing are quite evident in (a) and (b).

Sharpening Fourier Domain Filters

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Results of Butterworth High pass Filters



a b c

FIGURE 4.25 Results of highpass filtering the image in Fig. 4.11(a) using a BHPF of order 2 with $D_0 = 15$, 30, and 80, respectively. These results are much smoother than those obtained with an ILPF.

Sharpening Fourier Domain Filters

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Results of Gaussian High pass Filters

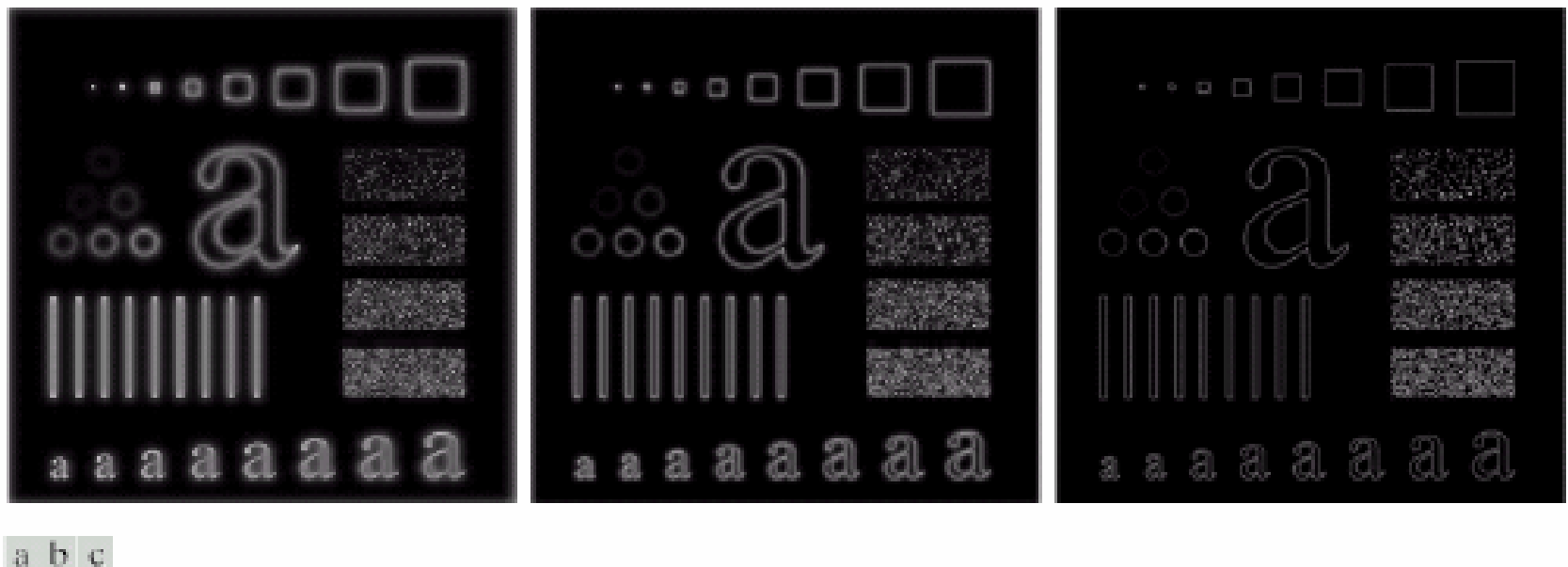


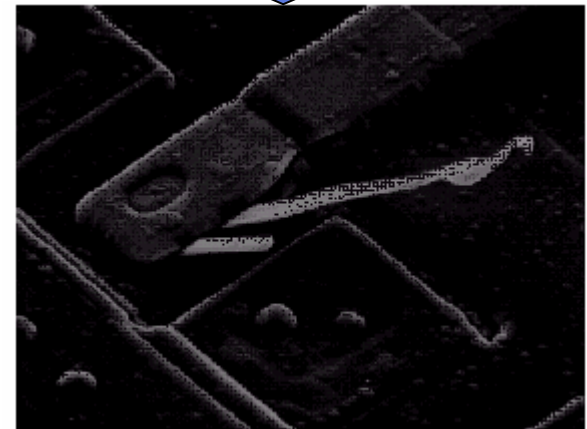
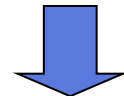
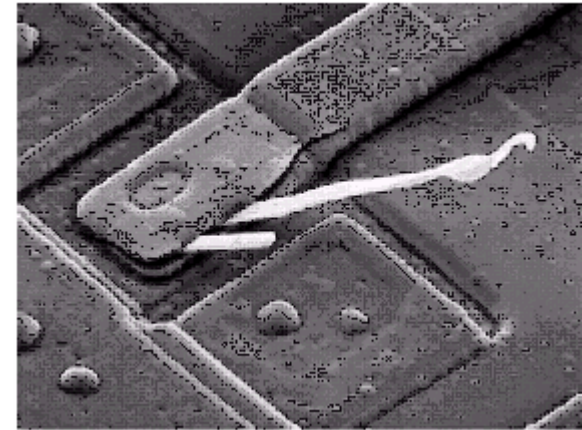
FIGURE 4.26 Results of highpass filtering the image of Fig. 4.11(a) using a GHPF with $D_0 = 15$, 30, and 80, respectively. Compare with Figs. 4.24 and 4.25.

Notch Filters

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- A filter which rejects or passes some frequencies
- This filter is to force the $F(0,0)$ which is the average value of an image (dc component of the spectrum) to zero

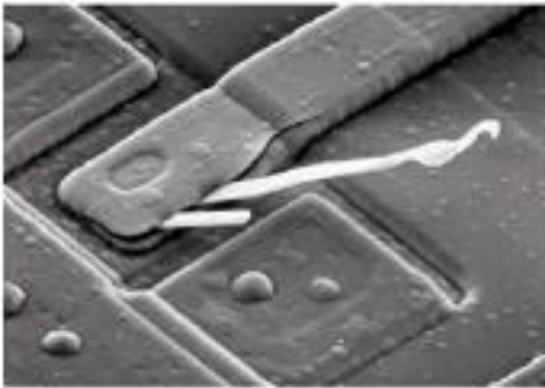
$$H(u, v) = \begin{cases} 0 & \text{if } (u, v) = (M/2, N/2) \\ 1 & \text{otherwise} \end{cases}$$



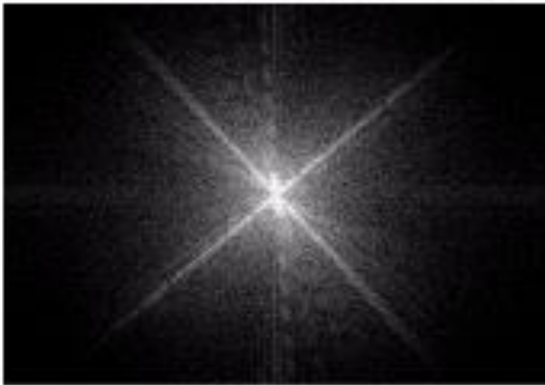
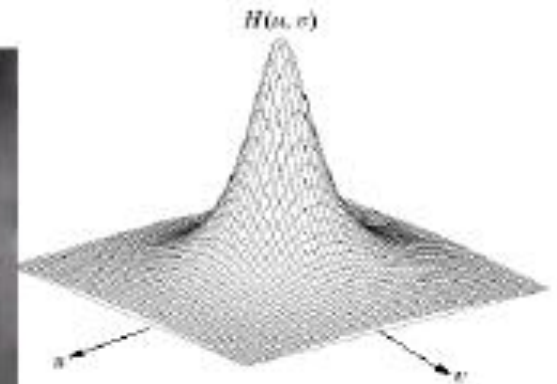
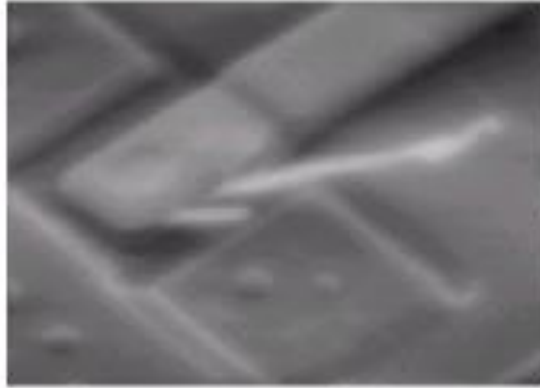
Low and High pass Filtering (Examples)

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Original image



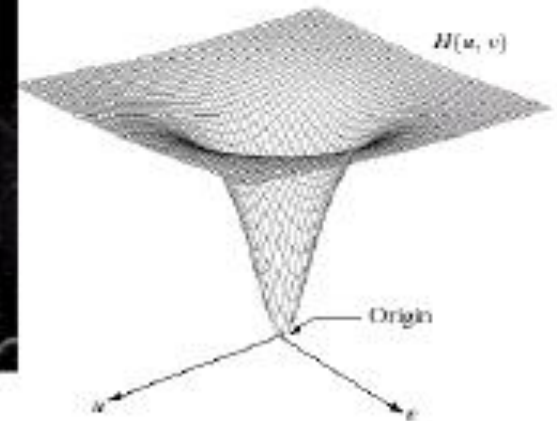
LPF applied



DFT of the image



HPF applied

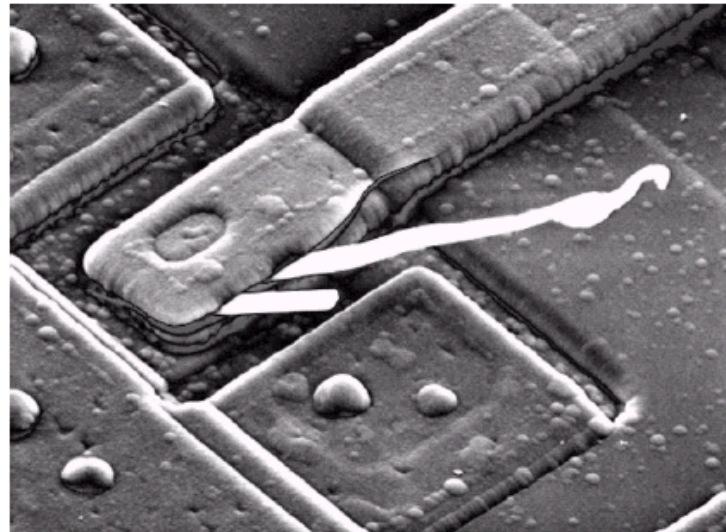


Add the $\frac{1}{2}$ of filter height to $F(0,0)$ of the high pass filter

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- ❑ As expected, the result after applying Gaussian LPF is a blurred image and the image after Gaussian HPF is sharp, with little smooth gray level detail because the $F(0,0)$ term has been set to zero
- ❑ In reality the average of the displayed image can't be zero as it needs to have negative gray levels, the output image needs to scale the gray level
- ❑ A procedure normally followed is to add a constant to the filter so that it will not completely eliminate $F(0,0)$

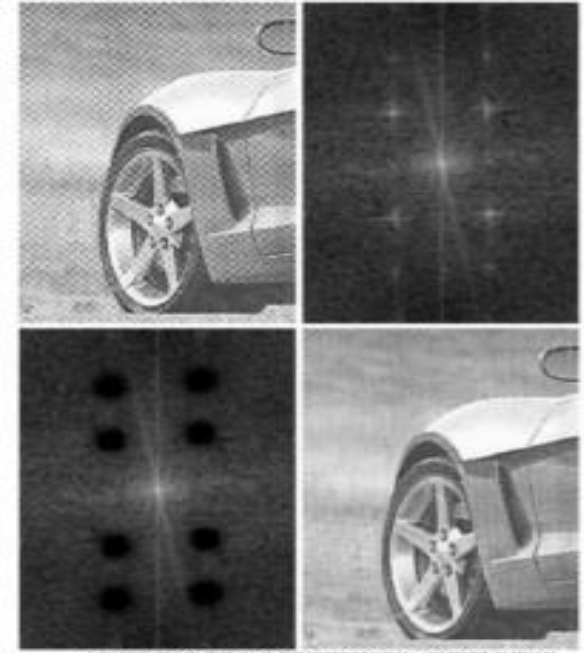
FIGURE 4.8
Result of highpass filtering the image in Fig. 4.4(a) with the filter in Fig. 4.7(c), modified by adding a constant of one-half the filter height to the filter function. Compare with Fig. 4.4(a).



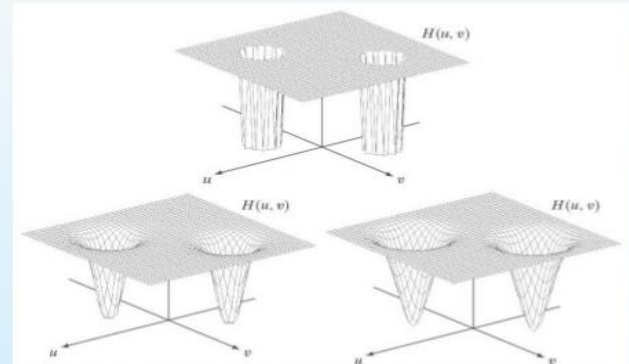
Notch Filters Steps

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- ❑ Look at spectrum $F(u,v)$ of noisy image $f(x,y)$ and find frequencies corresponding to noise
- ❑ Create a mask image $H(u,v)$ with notches at those places, 1's elsewhere
- ❑ Multiply mask with image transform; this zero outs the noise frequencies
- ❑ $G(u,v) = F(u,v) \times H(u,v)$
- ❑ Take inverse fourier transform of the restored image, $g(x,y) = \text{F}^{-1}(G(u,v))$



Some Notchreject Filters



References

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1. DIP by Gonzalez
2. http://en.wikipedia.org/wiki/Joseph_Fourier

For any query Feel Free to ask



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